

Chapter 1
The Foundations : Logic and Proofs
Kenneth H. Rosen 7th edition

Section 1.6 :Rules of Inference

Rules of Inference

- ▶ Infer means:
 - ▶ To deduce or conclude (something) from evidence and reasoning rather than from explicit statements.
- ▶ Proofs in mathematics are
 - ▶ Valid arguments
 - ▶ These arguments establish the truth of mathematical statements.
- ▶ By an *argument*, we mean a sequence of statements that end with a conclusion.
- ▶ By *valid*, we mean that the conclusion, or final statement of the *argument*, must follow from the truth of the preceding statements, or *premises* of the argument.
- ▶ An argument is valid if the truth of all its premises implies that the conclusion is true
 - ▶ i.e. the conclusion is true if the premises are all true.



Rules of Inference(Contd.)

- ▶ From the definition of a valid argument form we see that the argument form with premises $p_1, p_2, \dots, p_n$ and conclusion q is valid if and only if $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is a tautology.
- ▶ The key to showing that an argument in propositional logic is valid is to show that its argument form is valid.



Rules of Inference(Contd.)

<i>Rule of Inference</i>	<i>Name</i>	<i>Rule of Inference</i>	<i>Name</i>
p $p \rightarrow q$ $\therefore q$	<i>Modus ponens</i>	p $\therefore p \vee q$	<i>Addition</i>
$\neg q$ $p \rightarrow q$ $\therefore \neg p$		$p \wedge q$ $\therefore p$	
$p \rightarrow q$ $q \rightarrow r$ $\therefore p \rightarrow r$	<i>Hypothetical syllogism</i>	p q $\therefore p \wedge q$	<i>Conjunction</i>
$p \vee q$ $\neg p$ $\therefore q$		$p \vee q$ $\neg p \vee r$ $\therefore q \vee r$	
	<i>Disjunctive syllogism</i>		<i>Resolution</i>



Rules of Inference(Contd.)

▶ **Example 1:**

Consider the following argument:

- ▶ “If you have a current password, then you can log onto the network.”
- ▶ “You have a current password.”
- ▶ Therefore,
- ▶ “You can log onto the network.”



Rules of Inference(Contd.)

► **Solution:**

► Let,

$p = \text{"You have a current password."}$

$q = \text{"You can log onto the network."}$

Then, the argument has the form

$$\frac{p \rightarrow q \quad p}{\therefore q}$$

Here, we can see that the argument uses the form of Modus Ponens rule. The argument is thus a valid one. Now if, both $p \rightarrow q$ and p are *true*, then q must also be true.



Rules of Inference(Contd.)

- ▶ What if, you have two premises, $p \rightarrow q$ and p and the conclusion as q where, not both of the premises are true?



Rules of Inference(Contd.)

▶ **Example 2:**

Consider the following argument:

- ▶ “If you have access to the network, then you can change your grades.”
- ▶ “You have access to the network.”
- ▶ Therefore,
- ▶ “You can change your grades.”



Rules of Inference(Contd.)

► **Solution:**

► Let,

$p = \text{"You have access to the network."}$

$q = \text{"You can change your grades."}$

Then, the argument has the form

$$\frac{p \rightarrow q \quad p}{\therefore q}$$

Here, we can see that the argument uses the form of Modus Ponens rule. The argument is thus a valid one. Now, both $p \rightarrow q$ and q are not *true*, namely the first one is *false*. Thus, we cannot conclude that q is *true*.



Rules of Inference(Contd.)

▶ **Example 3:**

- ▶ Determine whether the argument given here is valid and determine whether its conclusion must be true because of the validity of the argument.

- ▶ If $\sqrt{2} > \frac{3}{2}$, then $(\sqrt{2})^2 > \left(\frac{3}{2}\right)^2$. We know that $\sqrt{2} > \frac{3}{2}$. Consequently $(\sqrt{2})^2 = 2 > \left(\frac{3}{2}\right)^2 = \frac{9}{4}$.



Rules of Inference(Contd.)

► **Solution:**

► Let,

$$p = \sqrt{2} > \frac{3}{2}$$

$$q = (\sqrt{2})^2 > \left(\frac{3}{2}\right)^2$$

The argument can be represented as

$$\frac{p \rightarrow q \quad p}{\therefore q}$$

The argument is valid as it is constructed using modus ponens. But, we cannot conclude that the conclusion is true. Because, the premises p is *false*. Also by observation, we can see that the conclusion, q is also *false*.



Rules of Inference(Contd.)

▶ **Example 4:**

- ▶ State which rule of inference is the basis of the following argument:

- ▶ *“It is below freezing now. Therefore, it is either below freezing or raining now.”*

▶ **Solution:**

- ▶ Let,

- $p = \text{“It is below freezing now.”}$

- $q = \text{“It is raining now.”}$

- The argument can be represented as,

$$\frac{p}{\therefore p \vee q}$$

This argument uses the addition rule.



Rules of Inference(Contd.)

▶ **Example 5:**

- ▶ State which rule of inference is used in the argument:
 - ▶ *If it rains today, then we will not have a barbecue today. If we do not have a barbecue today, then we will have a barbecue tomorrow. Therefore, if it rains today, then we will have a barbecue tomorrow.*

▶ **Solution:**

- ▶ Let,
 $p = \text{"It is raining today."}$
 $q = \text{"We will not have a barbecue today."}$
 $r = \text{"We will have a barbecue tomorrow."}$



Rules of Inference(Contd.)

The argument can be represented as,

$$\begin{array}{c} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

The argument thus uses the hypothetical syllogism rule.



Using Rules of Inference to Build Arguments

▶ **Example 1:**

▶ Show that the premises

- ▶ “It is not sunny this afternoon and it is colder than yesterday,”
- ▶ “We will go swimming only if it is sunny,”
- ▶ “If we do not go swimming, then we will take a canoe trip,”
- ▶ “If we take a canoe trip, then we will be home by sunset”

Lead to the conclusion

- ▶ “We will be home by sunset.”



Using Rules of Inference to Build Arguments(Contd.)

► **Solution:**

► Let,

$p = \text{"It is sunny this afternoon."}$

$q = \text{"It is colder than yesterday."}$

$r = \text{"We will go swimming."}$

$s = \text{"We will take a canoe trip."}$

$t = \text{"We will be home by sunset."}$

Then, the premises become,

1. $\neg p \wedge q$

2. $r \rightarrow p$

3. $\neg r \rightarrow s$

4. $s \rightarrow t$

The conclusion is simply t .



Using Rules of Inference to Build Arguments(Contd.)

We construct an argument to show that our premises lead to the desired conclusion as follows,

<i>Steps</i>	<i>Reasons</i>
1. $\neg p \wedge q$	<i>premise</i>
2. $\neg p$	<i>Simplification using (1)</i>
3. $r \rightarrow p$	<i>premise</i>
4. $\neg r$	<i>Modulus tollens using (2)and (3)</i>
5. $\neg r \rightarrow s$	<i>premise</i>
6. s	<i>Modulus ponens using (4)and (5)</i>
7. $s \rightarrow t$	<i>premise</i>
8. t	<i>Modulus ponens using (6)and (7)</i>

Thus, we can see that our premises lead to the desired conclusion.



Using Rules of Inference to Build Arguments(Contd.)

▶ **Example 2:**

▶ Show that the premises

- ▶ “If you send me an e-mail message, then I will finish writing the program.”
- ▶ “If you do not send me an e-mail message, then I will go to sleep early.”
- ▶ “If I go to sleep early, then I will wake up feeling refreshed.”

▶ Lead to the conclusion

- ▶ “If I do not finish writing the program, then I will wake up feeling refreshed.”



Using Rules of Inference to Build Arguments(Contd.)

► **Solution:**

► Let,

p = “*You send me an e – mail message.*”

q = “*I will finish writing the program.*”

r = “*I will go to sleep early.*”

s = “*I will wake up feeling refreshed.*”

Then the premises become,

1. $p \rightarrow q$

2. $\neg p \rightarrow r$

3. $r \rightarrow s$

The conclusion is $\neg q \rightarrow s$



Using Rules of Inference to Build Arguments(Contd.)

We construct an argument to show that our premises lead to the desired conclusion as follows,

<i>Steps</i>	<i>Reasons</i>
1. $p \rightarrow q$	<i>premise</i>
2. $\neg q \rightarrow \neg p$	<i>Contrapositive rule</i>
3. $\neg p \rightarrow r$	<i>premise</i>
4. $\neg q \rightarrow r$	<i>Hypothetical syllogism using (3)</i>
5. $r \rightarrow s$	<i>premise</i>
6. $\neg q \rightarrow s$	<i>Hypothetical syllogism using (3)</i>

Thus, we can see that our premises lead to the desired conclusion.



Resolution

- ▶ Resolution is nothing but a rule of inference based on the tautology,

$$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$$

- ▶ Using this resolution, we can derive rule of inference.

- ▶ Let,

r = False.

Then the resolution becomes,

$$((p \vee q) \wedge \neg p) \rightarrow q$$

This is the same as disjunctive syllogism.



Resolution(Contd.)

- ▶ **Example 1:**
- ▶ Use resolution to show that the hypotheses
 - ▶ “Jasmine is skiing or it is not snowing”
 - ▶ “It is snowing or Bart is playing hockey”
- ▶ Imply that
 - ▶ “Jasmine is skiing or Bart is playing hockey.”



Resolution(Contd.)

► **Solution:**

► Let,

$p = \text{"It is snowing"}$

$r = \text{"Jasmine is skiing"}$

$q = \text{"Bart is playing hockey."}$

The hypotheses can be represented as follows

1. $\neg p \vee r$

2. $p \vee q$

The resolution suggests that

$$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$$

Thus, the hypotheses implies,

$$(q \vee r)$$



Resolution(Contd.)

- ▶ **Example 2:**
- ▶ Show that the premises $(p \wedge q) \vee r$ and $r \rightarrow s$ imply the conclusion $p \vee s$.



Resolution(Contd.)

► **Solution:**

- We can rewrite the premises $(p \wedge q) \vee r$ as two clauses, $p \vee r$ and $q \vee r$. We can also replace $r \rightarrow s$ by the equivalent clause $\neg r \vee s$. Using the two clauses $p \vee r$ and $\neg r \vee s$, we can use resolution to conclude $p \vee s$.



Rules of Inference for Quantifiers

- ▶ Like the rules of inference for propositions, we now we see the rules of inference for quantified statements.

<i>Rules of Inference</i>	<i>Name</i>
$\frac{\forall xP(x)}{\therefore P(c)}$	<i>Universal instantiation</i>
$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall xP(x)}$	<i>Universal generalization</i>
$\frac{\exists xP(x)}{\therefore P(c) \text{ for some element } c}$	<i>Existential instantiation</i>
$\frac{P(c) \text{ for some element } c}{\therefore \exists xP(x)}$	<i>Existential generalization</i>



Rules of Inference for Quantifiers(Contd.)

▶ Universal Instantiation

- ▶ $P(c)$ is true, where c is a particular member of the domain, given the premise $\forall xP(x)$.

▶ Universal Generalization

- ▶ $\forall xP(x)$ is true, given the premise that $P(c)$ is true for all elements c in the domain.
- ▶ We show that $\forall xP(x)$ is true by taking an arbitrary element c from the domain and showing that $P(c)$ is true.
- ▶ The element c that we select must be an arbitrary, and not a specific, element of the domain.



Rules of Inference for Quantifiers(Contd.)

▶ Existential Instantiation

- ▶ Allows us to conclude that there is an element c in the domain for which $P(c)$ is true if we know that $\exists xP(x)$ is true.
- ▶ We cannot select an arbitrary value of c here, but rather it must be a c for which $P(c)$ is true.

▶ Existential Generalization

- ▶ Allows us to conclude that $\exists xP(x)$ is true when a particular element c with $P(c)$ true is known.
- ▶ That is, if we know one element c in the domain for which $P(c)$ is true, then we know that $\exists xP(x)$ is true.



Rules of Inference for Quantifiers(Contd.)

- ▶ **Example 1:**
- ▶ Show that the premises
 - ▶ “Everyone in this discrete mathematics class has taken a course in computer science”
 - ▶ “Marla is a student in this class”
- ▶ Imply the conclusion
 - ▶ “Marla has taken a course in computer science.”



Rules of Inference for Quantifiers(Contd.)

► **Solution:**

► Let,

$D(x)$ = “ *x is in this Discrete Mathematics class.*”

$C(x)$ = “ *x has taken a course in Computer Science.*”

Then, the premises can be represented as,

1. $\forall x(D(x) \rightarrow C(x))$
2. $D(Marla)$

The conclusion is simply, $C(Marla)$.



Rules of Inference for Quantifiers(Contd.)

We construct an argument to show that our premises lead to the desired conclusion as follows,

<i>Steps</i>	<i>Reasons</i>
1. $\forall x(D(x) \rightarrow C(x))$	<i>premise</i>
2. $D(\text{Marla}) \rightarrow C(\text{Marla})$	<i>Universal Instantiation using (1)</i>
3. $D(\text{Marla})$	<i>premise</i>
4. $C(\text{Marla})$	<i>Modus ponens using (2) and (3)</i>

Thus, we can see that our premises lead to the desired conclusion.



Rules of Inference for Quantifiers(Contd.)

- ▶ **Example 2:**
- ▶ Show that the premises
 - ▶ “A student in this class has not read the book”
 - ▶ “Everyone in this class passed the first exam”
- ▶ Imply the conclusion
 - ▶ “Someone who passed the first exam has not read the book.”



Rules of Inference for Quantifiers(Contd.)

► **Solution:**

► Let,

$C(x)$ = “*x is in this class.*”

$B(x)$ = “*x has read the book.*”

$P(x)$ = “*x passed the first exam.*”

Then the premises can be represented as,

1. $\exists x(C(x) \wedge \neg B(x))$
2. $\forall x(C(x) \rightarrow P(x))$

The conclusion is simply, $\exists x(P(x) \wedge \neg B(x))$.



Rules of Inference for Quantifiers(Contd.)

We construct an argument to show that our premises lead to the desired conclusion as follows,

<i>Steps</i>	<i>Reasons</i>
1. $\exists x(C(x) \wedge \neg B(x))$	<i>premise</i>
2. $C(a) \wedge \neg B(a)$	<i>Existential instantiation using (1)</i>
3. $C(a)$	<i>Simplification from (2)</i>
4. $\neg B(a)$	<i>Simplification from (2)</i>
5. $\forall x(C(x) \rightarrow P(x))$	<i>premise</i>
6. $C(a) \rightarrow P(a)$	<i>Universal instantiation from (5)</i>
7. $P(a)$	<i>Modus ponens from (3) and (6)</i>
8. $P(a) \wedge \neg B(a)$	<i>Conjunction from (4) and (7)</i>
9. $\exists x(P(x) \wedge \neg B(x))$	<i>Existential generalization from (8)</i>

Thus, we can see that our premises lead to the desired conclusion.



THE END

